The multi-species ancestral recombination graph reveals admixture dynamics in a hybrid baboon population

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Introduction

Many primate lineages naturally hybridize but remain taxonomically and genetically distinct. However, **no study has yet identified genetic isolating barriers that prevent free gene flow**.

Here, we developed new methods for inferring the coalescent history of introgressed regions, generated 4 new diploid near-telomere-to-telomere (T2T) genome assemblies, and used them to study a natural baboon hybrid zone in southern Kenya. This population is composed of both "recent hybrids" (descendants of gene flow in the last ~8 generations) and "historical hybrids" (descendants of gene flow hundreds of generations in the past), allowing us to contrast patterns of introgression in early and later phases of selection.

We hypothesized that information about coalescent timing of introgressed regions would be more informative than local ancestry information alone, such that:

- "Shallow" coalescent regions with a common ancestor in the minor parent (anubis baboons) post-split would be preferentially purged over time.
- Regions of the baboon genome with a high proportion of "shallow" coalescent trees would be more strongly correlated with proxies for selection, such as F_{ST}, local recombination rate, and local genetic diversity.
- Admixture mapping of a key fitness-related trait, 1st year offspring survival, would identify regions where anubis ancestry is deleterious, which co-localize with purging of "shallow" introgressed regions.

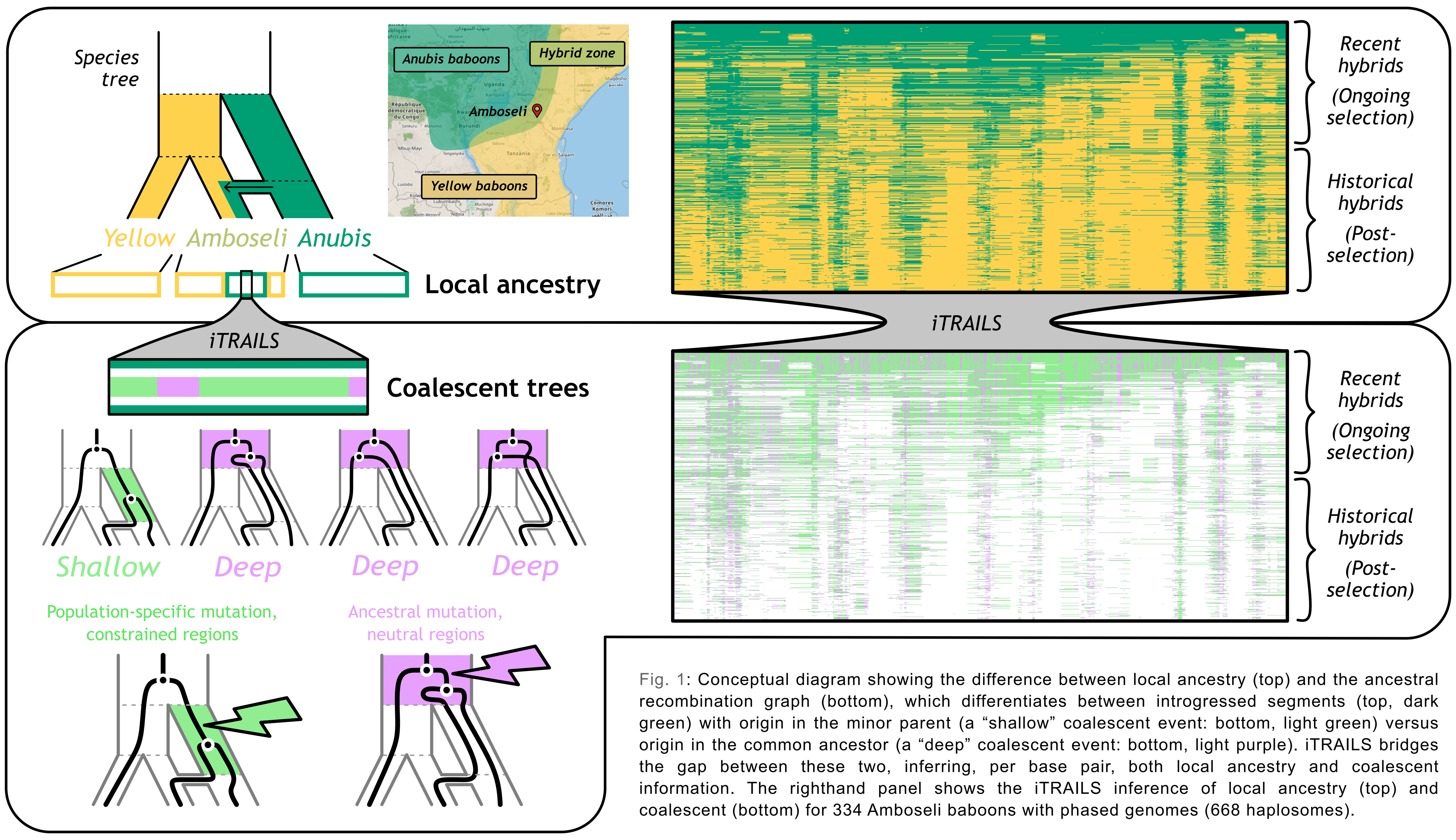


Fig. 1: Conceptual diagram showing the difference between local ancestry (top) and the ancestral recombination graph (bottom), which differentiates between introgressed segments (top, dark green) with origin in the minor parent (a "shallow" coalescent event: bottom, light green) versus origin in the common ancestor (a "deep" coalescent event: bottom, light purple). iTRAILS bridges the gap between these two, inferring, per base pair, both local ancestry and coalescent information. The righthand panel shows the iTRAILS inference of local ancestry (top) and coalescent (bottom) for 334 Amboseli baboons with phased genomes (668 haplosomes).

Results

Regions that are strongly diverged between the anubis and yellow parents (based on data from putatively unadmixed reference populations) have lower introgressed anubis ancestry in historical Amboseli hybrids (Fig. 2A).

This pattern is driven entirely by regions of the anubis genome where "shallow" coalescent trees are most common (Fig. 2B), in support of the hypothesis that these "shallow" regions are the primary target of selection against introgression.

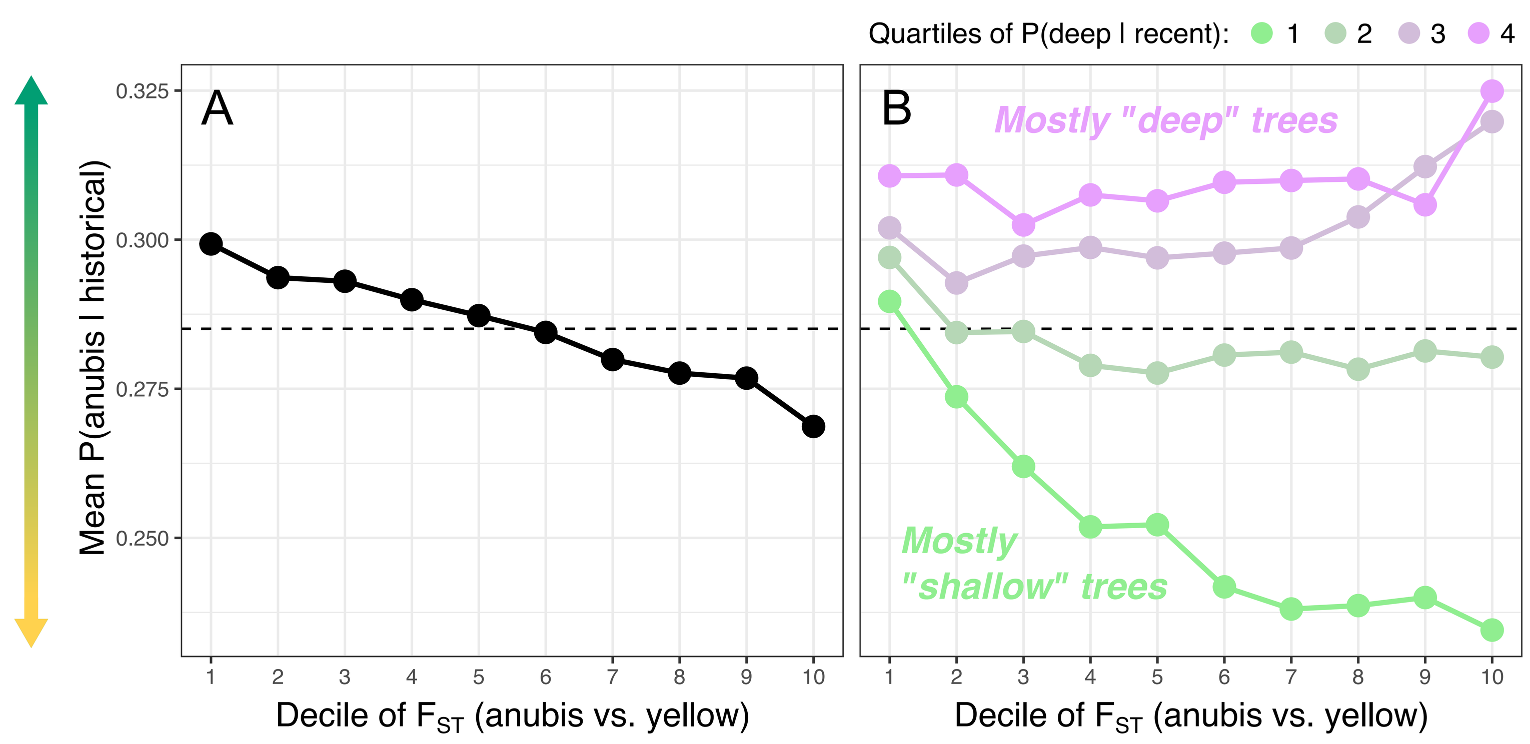
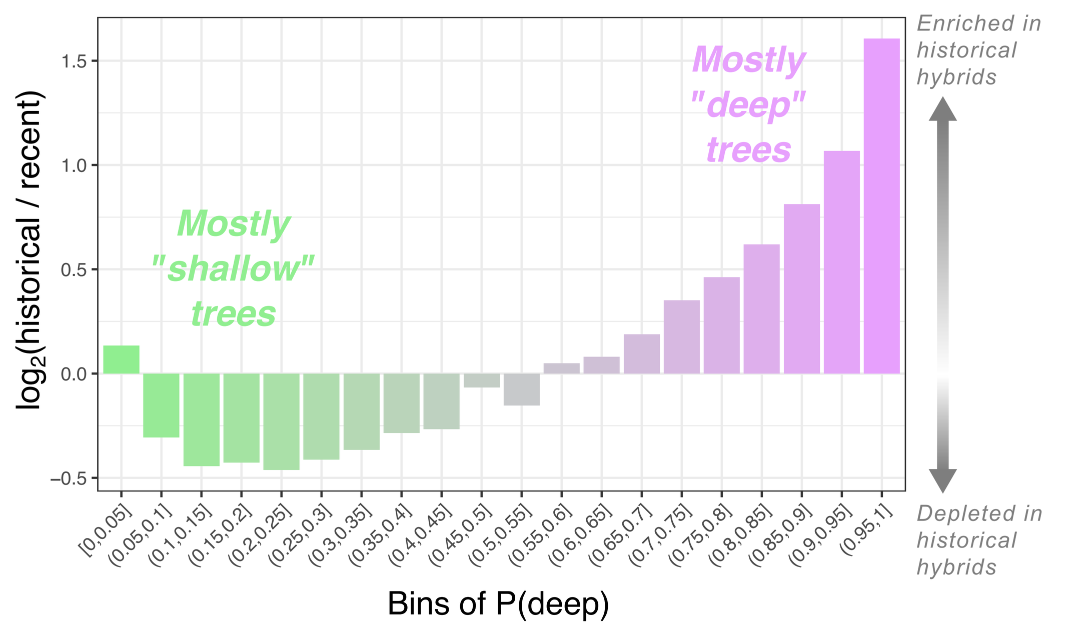


Fig. 2: The F_{ST} between unadmixed anubis and yellow baboons plotted against the proportion of anubis ancestry in historical hybrids, or P(anubis | historical). The lefthand plot (A) shows the average P(anubis | historical) in F_{ST} deciles, while the righhand plot (B) shows the same, but further segregated by quartiles of the proportion of deep coalescent in anubis fragments of recent hybrids, or P(deep | recent).

Comparing recent hybrids to historical hybrids shows that "shallow" trees are preferentially purged genome wide compared to "deep" trees, within segments of anubis local ancestry.



Regions with >75% deep ancestry increase by 1.3-to-3 fold following many generations of selection, while regions with <25% deep ancestry decrease by around 1.3-fold.

Fig. 3: Fold change, in $\log_2$ scale, in the distribution of P(deep) in historical vs. recent hybrids.

To test whether coalescent information helps identify genetic isolating barriers, we drew on >50 years of behavioral and demographic data from the Amboseli baboons to identify regions of the genome where local ancestry of mothers predicts the survival of their offspring to age 1 (n=990 offspring from 201 unique mothers).

Controlling for maternal rank, maternal loss in the 1st year of life, maternal global anubis ancestry, and maternal ID, we identify a highly polygenic architecture for ancestry effects on offspring survival (Fig. 4A).

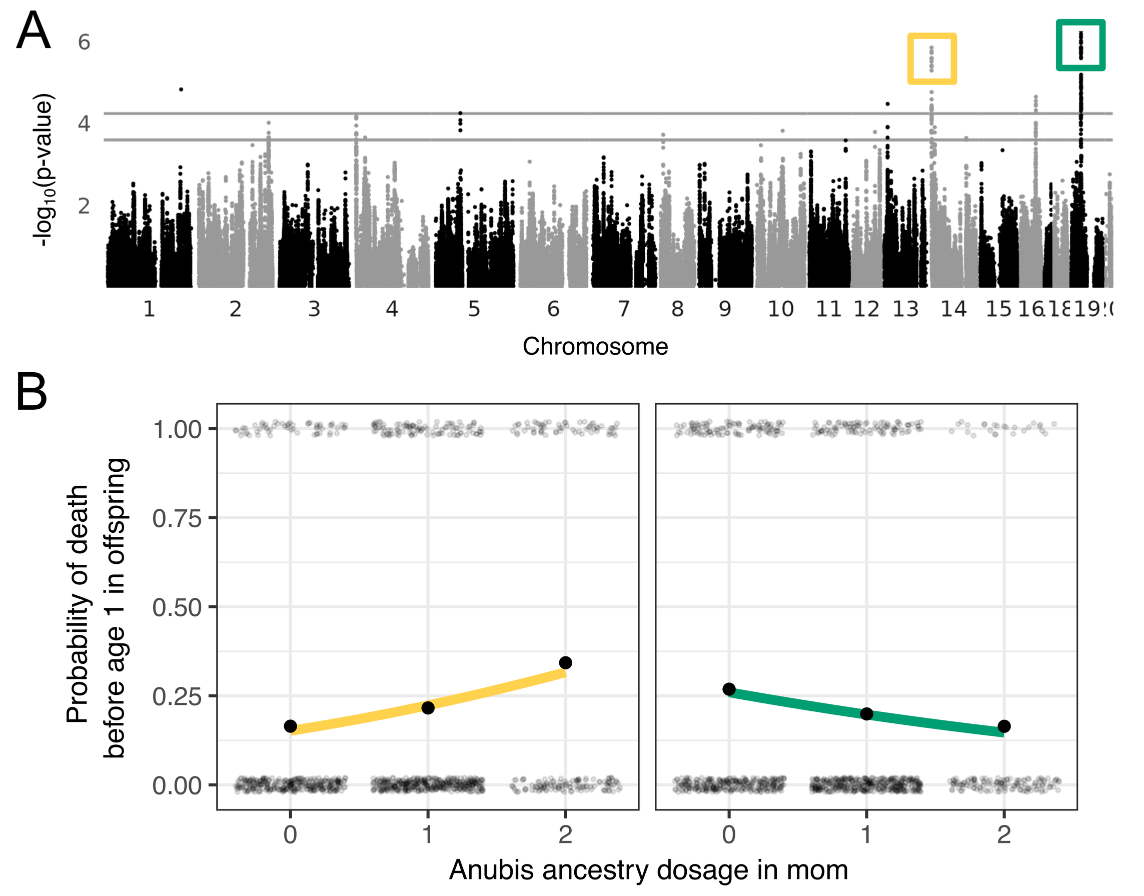
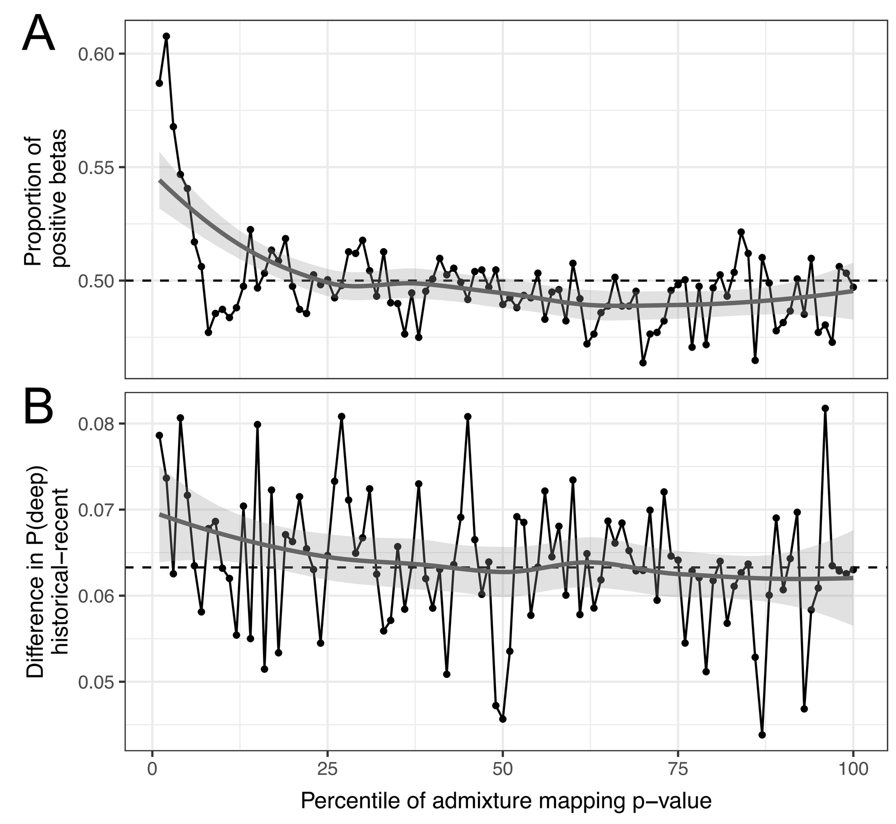


Fig. 4: (A) Manhattan plot of the admixture mapping p-values (horizontal lines show significance thresholds at 0.25 (bottom) and 0.1 (top) false discovery rates). The two most significant peaks are highlighted in yellow and green, and (B) shows the relationship between local maternal ancestry and the probability of offspring survival for each of these two peaks (logistic regression). The yellow curve shows a locus where local maternal yellow ancestry predicts better offspring survival, while the green curve shows a case where local maternal anubis ancestry predicts better offspring survival.

Consistent with overall selection against introgressed anubis ancestry, the strongest supported admixture mapping peaks are biased towards a positive effect (reduced offspring survival for locally anubis-like mothers) (Fig. 5A).

In the strongest supported admixture mapping peaks, "shallow" coalescent trees are purged more strongly than in the rest of the genome, based on comparisons between recent and historical Amboseli hybrids (Fig. 5B).



The purging pattern is driven entirely by the subset of admixture mapping peaks with positive effects (reduced offspring survival for locally anubis-like mothers), not negative effects (reduced offspring survival for locally yellow-like mothers) (Fig. 5C).

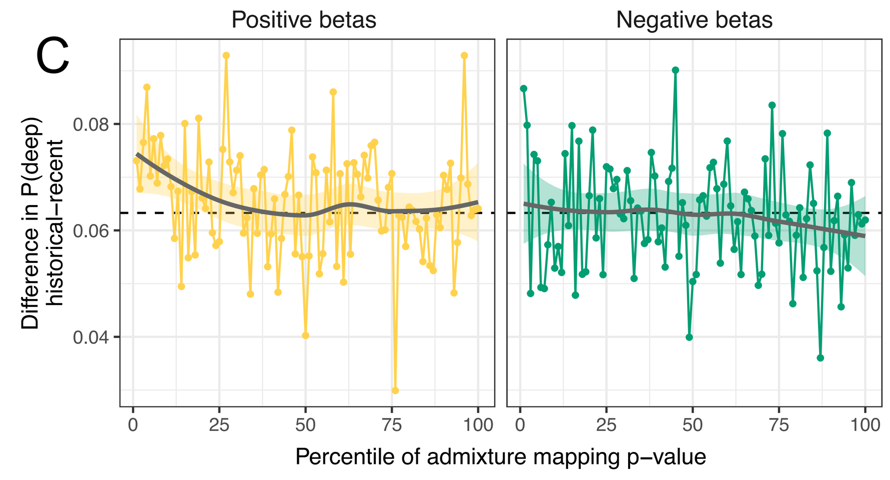


Fig. 5: Relationship between the admixture mapping p-values and the proportion of positive betas (A), and between the proportion of positive betas and the difference in P(deep) in historical vs. recent hybrids (B). Panel C is the same as panel B, but segregated by the sign of the betas.

Discussion

- In the complex hybrid zones typical of many primates (including our own lineage), single major effect genetic isolating loci are likely to be rare.
- Our results reinforce the **polygenic nature of reproductive isolation** in a natural primate hybrid zone and demonstrate the power of a **multispecies coalescent approach** to refine the targets of selection against introgression, beyond traditional local ancestry approaches.
- Further, by integrating field data from the same hybrid population, we were able to establish a **direct link between a fitness-relevant trait and the purging of introgressed variation**.
- This combined strategy provides a roadmap for how population genomic and statistical genetic approaches can be integrated with data from long-term field studies to investigate divergence and speciation in the primate lineage.

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